

The core strategy matrix

Quantifier	Goal: Prove True	Goal: Prove False
$\forall x$ ("For all x ")	Algebraic proof or proof by cases (must work for an arbitrary variable)	Provide one counterexample where the condition breaks
$\exists x$ ("There exists an x ")	Provide one working example that satisfies the condition	Algebraic impossibility proof (show the opposite is universally true)

1. Handling Equations ($=$)

- For $\forall x(\text{LHS} = \text{RHS})$:
 - If True: Simplify both sides using algebraic identities until you reach an identity like $0 = 0$ or $x = x$ (e.g., $(-x)^2 = x^2$). Never just test numbers.
 - If False: Pick a single value for x where the sides differ (e.g., for $\forall x(x + 1 = x)$, testing $x = 0$ gives $1 \neq 0$).
- For $\exists x(\text{LHS} = \text{RHS})$:
 - If True: Solve the equation algebraically for x , then verify that the solution lives in your allowed domain (e.g., $x^2 = 2 \implies x = \sqrt{2} \in \mathbb{R}$).
 - If False: Solve the equation and show that every solution falls outside the domain (e.g., $x^2 = -1 \implies x = \pm i \notin \mathbb{R}$, or $x^2 = 2 \implies x = \pm\sqrt{2} \notin \mathbb{Z}$).

2. Handling Inequalities ($<$, $>$, $\leq$, $\geq$)

- For $\forall x(\text{Inequality})$:
 - If True:
 - Standard properties: Start from a known baseline, like $x^2 \geq 0$ or $|x| \geq 0$, and perform valid inequalities operations on both sides (e.g., adding 2 gives $x^2 + 2 \geq 2 > 1$).
 - Proof by Cases: Split the domain into logical partitions (e.g., $x > 0$, $x = 0$, $x < 0$) and show the statement holds in every case.
 - If False: Solve the inequality algebraically to find its valid range. Any value outside that valid range is an immediate counterexample (e.g., solving $2x > x$ yields $x > 0$, so $x = -1$ serves as a counterexample).
- For $\exists x(\text{Inequality})$:
 - If True: Find any single number in the domain that makes the inequality true (e.g., for $\exists x(2x > x)$, testing $x = 5$ gives $10 > 5$, which completes the proof).

- If False: Show that the inequality violates a mathematical rule for all numbers in the domain (e.g., showing $\exists x(x^2 < 0)$ is false because $x^2 \geq 0$ holds universally for all real numbers).

To quickly check if a statement is true or false on an exam before you write out the formal proof, you should use a two-step mental routine: Test the Trick Numbers, then Solve the Algebra.

Here is how to do it on your scratch paper:

Step 1: Test the "Trick Numbers" (The Guessing Phase)

Your first instinct should always be to plug in numbers that behave weirdly in math. Test these four specific numbers in your head:

1. 0 (Destroys multiplication and makes squares equal zero)
2. -1 (Flips signs and tests negative cases)
3. 1 (The baseline positive number)
4. 0.5 (Fractions get smaller when squared: $0.5^2 = 0.25$)

How this gives you the answer:

- If you are checking a $\forall$ (For All) statement and even one of these numbers makes the inequality fail, stop checking. The statement is False. (You now have your counterexample).
- If you are checking an $\exists$ (There Exists) statement and even one of these numbers makes the inequality work, stop checking. The statement is True. (You now have your proof example).

Step 2: Solve for x (The Confirmation Phase)

If plugging in numbers didn't give you an immediate answer, ignore the $\forall$ or $\exists$ symbol for a second and just solve the inequality like a normal algebra problem.

Let's say your inequality is $2x > x$. If you subtract x from both sides, you get the solution: $x > 0$.

Now look at that solution ($x > 0$) and ask yourself two questions based on your quantifier:

If the problem asks $\forall x(2x > x)$:

- Ask yourself: Does " $x > 0$ " cover every single number in the domain?
- Answer: No, it excludes negative numbers and zero.
- Conclusion: False.

If the problem asks $\exists x(2x > x)$:

- Ask yourself: Does " $x > 0$ " cover at least one number in the domain?

- Answer: Yes, there are plenty of numbers greater than zero.
- Conclusion: True.

Step-by-Step Decision Checklist

1. Check the Domain: Always identify if variables belong to integers ($\mathbb{Z}$) or real numbers ($\mathbb{R}$).
2. Form a Hypothesis: Test quick mental values (0, 1, -1 , fractions, or large numbers).
3. Select the Proof Type:
 - Proving $\forall$ true $\rightarrow$ Arbitrary algebra / Cases.
 - Proving $\forall$ false $\rightarrow$ One counterexample.
 - Proving $\exists$ true $\rightarrow$ One valid candidate.
 - Proving $\exists$ false $\rightarrow$ Show the negation is universally true.